MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, CYLINDER HEAD, 0°C TO 300°C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one type of single pointer, electrical resistance type temperature indicator.

1.2 Classification. The indicators shall be classified as follows:

MS28004-1	Fluorescent Dial Markings
MS28004-3	Lusterless White Dial Markings

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods Of
DOD-D-1000	Drawing, Engineering and Associated List
MIL-F-6836	Flanges, Mounting, Attachable, Aircraft Instrument
MIL-I-7070	Indicator, Temperature, Electrical-Resistance, Hermetically Sealed, General Specification For

STANDARDS

FEDERAL

FED-STD-595	Colors

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-794	Parts and Equipment, Procedures For Packaging
MS28004	Indicator-Temperature, Electrical Resistance, Cylinder Head, 0-300°C
MS33558	Numeral and Letter, Aircraft Instrument Dial, Standard Form Of
MS33638	Case, Instrument, Flange-Mounted, Aircraft

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

* ASTM D 3951 Packaging, Commercial

* (Application for copies of ASTM publications should be addressed to the American Society for Testing and Materiels, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

* 2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The indicator furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.4).

3.2 General. The requirements of MIL-I-7070 are applicable as requirements of this specification with the exceptions and additions specified herein. In the event of conflict, this specification shall govern.

3.3 Construction.

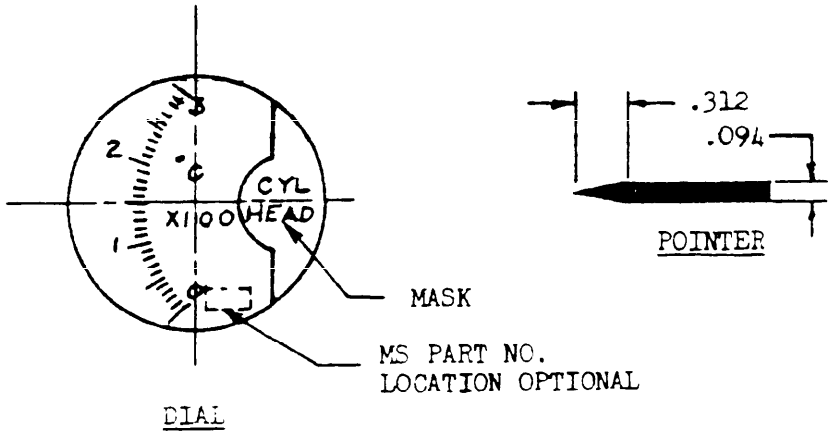
3.3.1 Case. The indicator case shall conform to MS28004.

3.3.2 Dial. The dial shall be as shown on Figure 1.

3.3.2.1 Dial markings. The dial markings shall be in accordance with MS33558 and Figure 1. Dimension of markings shall be as shown in the table in Figure 1. The lusterless black shall be color 37038 of FED-STD-595. The lusterless white shall be color 37875 of FED-STD-595.

3.3.2.2 Scale length. The scale of the indicator shall be as long as practicable and shall not be less than 1 3/4 inches as measured on the outer arc of the shortest graduation.

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MARKING	HEIGHT OR LENGTH ±.005	WIDTH OF LINE OR GRADUATIONS ±.005	MATERIAL OR FINISH
ALL NUMERALS	.156	.020	FLUORESCENT
MAJOR GRADUATION	.156	.031	LUMINESCENT
MINOR GRADUATION	.093	.020	MATERIAL
LETTERING "CX100"	.125	.020	OR
LETTERING "CYL HEAD"	.125	.020	LUSTERLESS
POINTER (VISIBLE PORTION)	----	----	WHITE
BACKGROUND OF DIAL (MS PART NO.)	---- .062	---- .010	LUSTERLESS BLACK

DIMENSIONS IN INCHES.

FIGURE 1. DIAL, POINTER, AND MARKINGS

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3.3.3 Pointer. The pointer shall be as shown on Figure 1. The shaded portion of the pointer shall be finished lusterless white, color 37875 of FED-STD-595. The unshaded portion shall be finished lusterless black, color 37038 of FED-STD-595.

3.3.3.1 Pointer overswing. The pointer overswing shall be not less than 2 angular degrees beyond each end of scale.

3.3.4 Electrical receptacle. An electrical receptacle shall be provided and located as shown on MS28004. The receptacle shall be wired as follows:

Terminal	Connection
A	+24V Battery
B	Resistance Bulb
C	Common-Negative 24V Battery and remaining bulb connection.

3.4 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawings requirements of DOD-D-1000 shall govern changes in the manufacturer's part number.

3.5 Weight. The weight of the completed indicator shall not exceed 0.75 pounds.

3.6 Identification of product. Indicator assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.6.1 Nameplate. The nameplate shall be marked with the following information:

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, CYLINDER HEAD
MS PART NO.
STOCK NO.
MANUFACTURER'S PART NO.
MANUFACTURER'S SERIAL NO.
CONTRACT OR ORDER NO.
MANUFACTURER'S NAME OR TRADE-MARK
US PROPERTY

* 4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General. The indicators shall be subjected to sampling, inspection and test procedures in accordance with MIL-I-7070, except as specified herein.

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4.3 Test voltage. The test voltage shall be 28.5 ±0.5 volts direct current (DC), unless otherwise specified.

4.3.1 Test voltage for the reduced voltage operation test shall be 22.5 ±.5 volts DC.

4.4 Test tolerances.

4.4.1 The tolerances, specified in Table I, not enclosed in parenthesis are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in Table I, enclosed in parenthesis, are applicable only when the indicators are subjected to a specific test independent of sequence.

4.4.2 Scale error at room temperature. The test resistance specified in Table I shall be applied to the resistance bulb terminals of the indicator. The test voltage shall be applied and the scale errors at the test points specified in Table I shall not exceed the tolerance specified in Table I.

4.4.3 Position error. When the indicator is subjected to the position error test, the maximum change in indication when voltage is applied shall not exceed 7° Centigrade (C).

4.4.4 Friction error. When the indicator is subjected to the friction error test, the maximum difference in reading before and after tapping shall not exceed 7°C.

4.4.5 Sealing. When the indicator is subjected to the sealing test no leaks shall result.

4.4.6 Insulation leakage. When the indicator is subjected to the insulation leakage test, the insulation resistance shall be a minimum of 20 megohms.

4.4.7 Reduced voltage operation. The indicator shall be subjected to a scale error at room temperature test with an applied voltage of 22.5 volts at the test points specified in Table I. The indications at the reduced voltage shall not differ from the indications with 28.5 volts applied by more than the tolerances specified in Table I.

4.4.8 Scale error at low temperature. When the indicator is subjected to the scale error at low temperature test, the scale errors shall not exceed the tolerances specified in Table I.

4.4.9 Scale error at high temperature. When the indicator is subjected to the scale error at high temperature test, the scale error shall not exceed the tolerances specified in Table I.

4.4.10 Magnetic effect. When the indicator is subjected to the magnetic effect test, it shall not cause the magnetic compass to deflect more than 5 degrees.

4.4.11 Resistance bulb current. A total resistance of 100 ohms shall be applied across the bulb-circuit terminals of the indicator, the test voltage applied and the bulb-circuit current measured. The current consumed in the resistance bulb circuit shall not exceed 22 milliamperes (ma).

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TABLE I - Test tolerances.

Test Point	Test Resistance	Scale Error at Room Temperature	Reduced Voltage	Scale Error at Low Temperature	Scale Error at High Temperature
		Tol ±	Tol ±	Tol ±	Tol ±
°C	OHms	°C	°C	°C	°C
0	50.00	5.0 (5.0)	4.0 (4.0)	10.0 (10.0)	10.0 (10.0)
+50	63.55	5.0 (5.0)	4.0 (4.0)	10.0 (10.0)	10.0 (10.0)
100	79.00	5.0 (5.0)	4.0 (4.0)	10.0 (10.0)	10.0 (10.0)
150	96.50	5.0 (5.0)	4.0 (4.0)	10.0 (10.0)	10.0 (10.0)
200	116.50	5.0 (5.0)	5.0 (5.0)	10.0 (10.0)	10.0 (10.0)
250	139.00	5.0 (5.0)	5.0 (5.0)	10.0 (10.0)	10.0 (10.0)
300	164.50	5.0 (5.0)	5.0 (5.0)	10.0 (10.0)	10.0 (10.0)

Test Point	Test Resist-ance	Scale Error at Room Tempera-ture After Open and Short Circuit	Scale Error at Room Temperature After Vibra-tion Failure	Extreme Low Temperature Exposure	
				Scale Error at Low Temperature	Scale Error at Room Temperature After Low Temperature Exposure
		Tol ±	Tol ±	Tol ±	Tol ±
°C	OHms	°C	°C	°C	°C
0	50.00	7.0 (7.0)	12.0 (12.0)	15.0 (10.0)	12.0 (5.0)
+50	63.55	7.0 (7.0)	12.0 (12.0)	15.0 (10.0)	12.0 (5.0)
100	79.00	7.0 (7.0)	12.0 (12.0)	15.0 (10.0)	12.0 (5.0)
150	96.50	7.0 (7.0)	12.0 (12.0)	15.0 (10.0)	12.0 (5.0)
200	116.50	7.0 (7.0)	12.0 (12.0)	15.0 (10.0)	12.0 (5.0)
250	139.00	7.0 (7.0)	12.0 (12.0)	15.0 (10.0)	12.0 (5.0)
300	164.50	7.0 (7.0)	12.0 (12.0)	15.0 (10.0)	12.0 (5.0)

Test Point	Test Resistance	Scale Error at Room Temperature After High Temperature Exposure	Scale Error at Room Tempera-ture After Salt Spray	Scale Error With Bulb Lead Resistance
		Tol ±	Tol ±	Tol ±
°C	OHms	°C	°C	°C
0	50.00	12.0 (5.0)	12.0 (5.0)	5.0 (5.0)
+50	63.55	12.0 (5.0)	12.0 (5.0)	5.0 (5.0)
100	79.00	12.0 (5.0)	12.0 (5.0)	5.0 (5.0)
150	96.50	12.0 (5.0)	12.0 (5.0)	5.0 (5.0)
200	116.50	12.0 (5.0)	12.0 (5.0)	5.0 (5.0)
250	139.00	12.0 (5.0)	12.0 (5.0)	5.0 (5.0)
300	164.50	12.0 (5.0)	12.0 (5.0)	5.0 (5.0)

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4.4.12 Open circuit and short circuit. When the indicator is subjected to the open circuit and short circuit test, the scale errors at room temperature after open and short circuit shall not exceed the tolerances specified in Table I.

4.4.13 Vibration error. When the indicator is subjected to the vibration error test, the maximum amplitude of pointer oscillation throughout the frequency range shall not exceed 7°C. The maximum difference between the original position of the pointer before vibration and position of the pointer during vibration shall not exceed 7°C. Pointer oscillation shall not be considered as pointer variation.

4.4.14 Vibration failure. When the indicator is subjected to the vibration failure test, the scale errors at room temperature after vibration shall not exceed the tolerances specified in Table I.

4.4.15 Response time. When the indicator is subjected to the response time test, the response time shall not exceed 3 seconds.

4.4.16 Damping factor. When the indicator is subjected to the damping factor test, the damping factor shall not be less than 1.6.

4.4.17 Power consumption. When the indicator is subjected to the power consumption tests, the power consumption shall not exceed 1.5 watts.

4.4.18 Extreme low temperature exposure. When the indicator is subjected to the extreme low temperature exposure test, the scale errors at low temperature shall not exceed the tolerances specified in Table I. The scale errors at room temperature after low temperature exposure shall not exceed the tolerances specified in Table I.

4.4.19 High temperature exposure. When the indicator is subjected to the high temperature exposure test, the scale errors at room temperature after high temperature exposure shall not exceed the tolerances specified in Table I.

4.4.20 Bulb lead resistance effect. The indicator shall be connected as shown in Figure 2 and subjected to the scale error at room temperature test. The indicator with bulb lead resistance shall not vary from the indications without bulb lead resistance by more than the tolerance specified in Table I.

4.4.21 Thermal shock. When the indicator is subjected to the thermal shock test, no damage to the indicator shall occur.

4.4.22 Salt spray. When the indicator is subjected to the sealing test after the salt spray test, no leaks shall result.

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking for shipment and storage shall be examined and tested to determine compliance with the requirements of Section 5, or the documents specified therein.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A, C or Commercial as specified (see 6.3).

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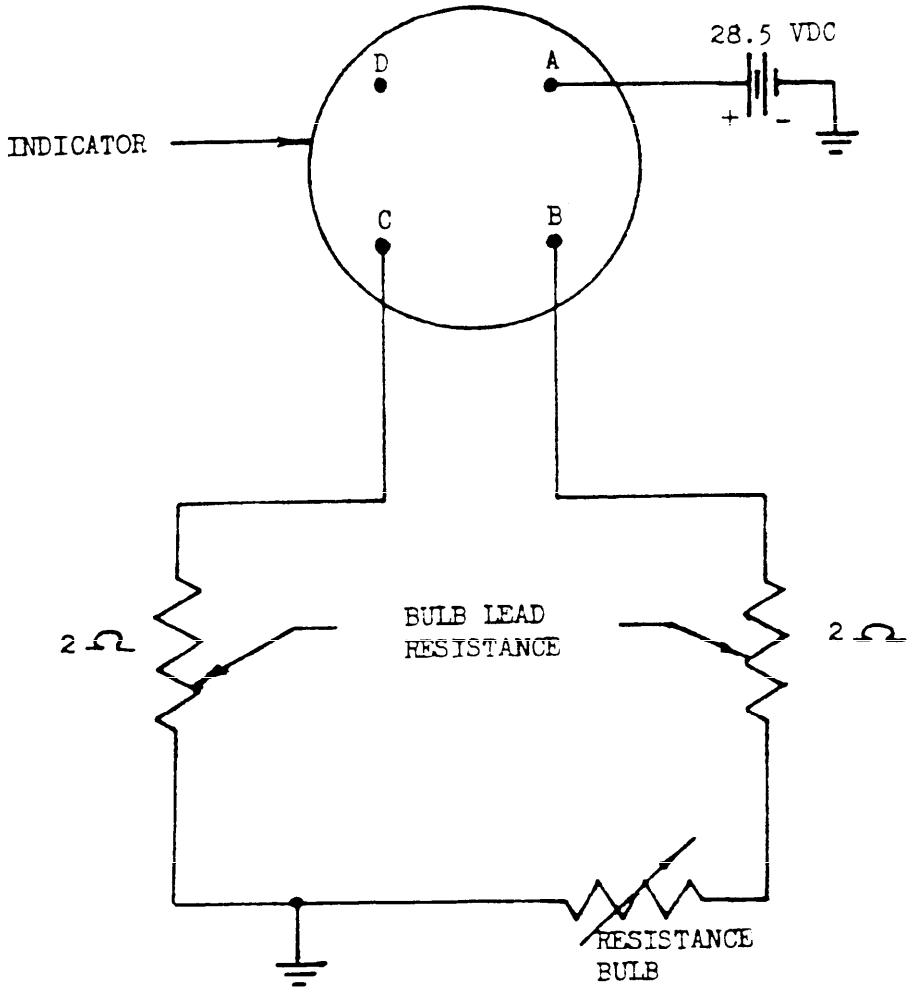


FIGURE 2. WIRING DIAGRAM

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* 5.1.1 Level A.

* 5.1.1.1 Cleaning. Indicators shall be cleaned in accordance with MIL-P-116.

* 5.1.1.2 Drying. Indicators shall be dried in accordance with MIL-P-116.

* 5.1.1.3 Preservation Applications. Preservatives shall not be used.

* 5.1.1.4 Packaging. Unless otherwise specified (see 6.3) each indicator shall be individually packaged in accordance with MIL-P-116, Method IC.

* 5.1.2 Level C. The Level C preservation for indicators shall conform to the requirements of MIL-STD-794 for this level.

* 5.1.3 Commercial. The commercial/industrial preservation of indicators shall be in accordance with the requirements of ASTM D 3951.

* 5.2 Packing. Packing shall be Level A, B, C or commercial, as specified (see 6.3).

* 5.2.1 Level A. The packaged indicators shall be packed in containers as specified in MIL-STD-794 for Level A protection.

* 5.2.2 Level B. The packaged indicators shall be packed in containers as specified in MIL-STD-794 for Level B protection.

* 5.2.3 Level C. The packaged indicators shall be packed in containers as specified in MIL-STD-794 for Level C protection.

* 5.2.4 Commercial. The packaged indicators shall be packed in accordance with the requirements of ASTM D 3951.

* 5.3 Marking.

* 5.3.1 Levels A, B and C. In addition to any special or other identification marking required by the contract (See 6.3), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

* 5.3.2 Commercial. Commercial/industrial marking shall be in accordance with the requirements of ASTM D 3951.

6. NOTES

6.1 Intended use. The MS28004 electrical resistance temperature indicator covered by this specification is intended to indicate the cylinder head temperature of reciprocating engines.

6.2 The MS28004 temperature indicator can be used on instrument panels requiring MS33638 case mountings by the application of an instrument mounting bezel in accordance with MIL-T-6836.

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6.3 Ordering data.

6.3.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type number of indicator desired.

c. Ordering data in accordance with MIL-I-7070.

d. Applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).

6.4 General. The instructions contained in Section 6 of MIL-I-7070 are applicable to and form a part of this specification except that information pertaining to qualification of products covered by this specification shall be obtained from:

Commander
San Antonio Air Logistic Center
Service Engineering Division
ATTN: MMEDO
Kelly Air Force Base, Texas 78241

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
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Army - ME

Preparing Activity:
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conducted on components which have been manufactured in accordance

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-7112B	Indicator, Temperature, Electrical Resistance, Cylinder Head, 0°C to 300°C

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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